THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM THREE
033 BIOLOGY
MARKING SCHEME 2026

SECTION A (16 MARKS)

Question 1

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
C	B	D	B	C	C	C	B	A	C

1 mark each = 10 marks

Question 2

(i)	(ii)	(iii)	(iv)	(v)	(vi)
B	E	D	G	A	C

1 mark each = 6 marks

SECTION B (54 MARKS)

Question 3:

(a) Any Four steps — **6 marks**

(i) Identification of a problem/question
(ii) Observation
(iii) Formulation of hypothesis
(iv) Planning/designing an experiment
(v) Identification of variables
(vi) Conducting the experiment
(vii) Collection/recording of data
(viii) Analysis of results
(ix) Drawing a conclusion
(x) Communicating/reporting findings .

(b) Importance of accurate recording **— 3 marks**

(i) Enables reliable analysis of results.
(ii) Makes comparison of results possible.
(iii) Helps another investigator repeat the experiment.
(iv) Reduces errors.
(v) Provides evidence for the conclusion.
(vi) Provides a permanent scientific record.

Total: 9 marks

Question 4:

(a) Poor personal hygiene and disease transmission — **6 marks**

(i) Failure to wash hands can transfer pathogens to food.

(ii) Poor hygiene allows microorganisms to multiply.

(iii) Improper disposal of human waste contaminates food/water.

(iv) Sharing contaminated personal items may transmit infections.

(v) Unclean surroundings provide breeding places for disease vectors.

(vi) Failure to maintain body cleanliness can encourage infections.

(b) Body protection against disease organisms, any three — **3 marks**

(i) Skin forms a physical barrier.

(ii) Mucus traps microorganisms.

(iii) Tears help wash away microorganisms.

(iv) Stomach acid destroys many pathogens.

(v) White blood cells destroy pathogens.

(vi) Antibodies help defend against specific pathogens.

Total: 9 marks

Question 5:

(a) Pathway of reflex action — 6 marks

(i) Stimulus detected by a receptor in the skin.

(ii) Impulse generated in the receptor.

(iii) Impulse travels along a sensory neurone.

(iv) Impulse passes through a relay neurone in the central nervous system/spinal cord.

(v) Impulse passes to a motor neurone.

(vi) Motor neurone carries impulse to the effector muscle, causing withdrawal.

(b) Distinctions — 3 marks

(i) Sensory vs motor neurone

- Sensory neurone carries impulses from receptor to CNS
- Motor neurone carries impulses from CNS to an effector

(ii) Voluntary vs involuntary movement

- Voluntary movement is consciously controlled
- Involuntary movement occurs automatically without conscious control

Total: 9 marks

Question 6:

(a) End products of digestion — 3 marks

(i) Starch → **glucose/simple sugars**

(ii) Proteins → **amino acids**

(iii) Fats → **fatty acids and glycerol**

(b) Adaptations of small intestine — 6 marks

1 mark each, any three:

(i) Very long/coiled, providing a large surface area.

(ii) Has numerous villi.

(iii) Villi have microvilli, increasing surface area.

(iv) Thin epithelial wall allows rapid absorption.

(v) Rich blood supply maintains concentration gradients.

(vi) Lacteals absorb fatty acids/glycerol products.

(vii) Suitable length allows sufficient time for absorption.

Total: 9 marks

Question 7:

(a) Adaptations of red blood cells — 3 marks

Any three:

(i) Biconcave shape gives a large surface area.

(ii) Thin shape provides a short diffusion distance.

(iii) Contain haemoglobin for oxygen transport.

(iv) Lack a nucleus, providing more space for haemoglobin.

(v) Flexible membrane allows passage through narrow capillaries.

(b)6 marks

Arteries	**Veins**
(i) Carry blood away from heart	(i) Carry blood towards heart
(ii) Thick, muscular, elastic walls	((ii)Thinner, less muscular walls
(iii)Usually no valves along main length	(iii) Have valves
(iv)Blood usually under high pressure	(iv) Blood usually under lower pressure
(v)Narrower lumen	(v) Wider lumen

Total: 9 marks

Question 8:

(a) Excretion and egestion — **4 marks**

Excretion: Removal of metabolic waste products from the body — 2 marks

Egestion: Removal of undigested/unabsorbed food from the alimentary canal — 1 mark

Difference: Excretion involves metabolic wastes, whereas egestion involves undigested food — 1 mark

(b) Urine formation in a nephron — **5 marks**

(i) Ultrafiltration: blood is filtered at the glomerulus/Bowman's capsule.

(ii) Water and small dissolved substances enter the nephron while large proteins and blood cells remain in blood.

(iii) Selective reabsorption occurs mainly in the proximal convoluted tubule.

(iv) Useful substances such as glucose, amino acids, appropriate amounts of salts and water are reabsorbed.

(v) Remaining fluid becomes urine and passes through collecting ducts → ureter → bladder → urethra.

Total:9 marks

SECTION C (30 MARKS)

Question 9:

Introduction: **(1 mark)**

(a) Characteristics **(6 Marks):**

Coniferophyta: Bear naked seeds in cones; Needle-like or scale-like leaves; Mostly evergreen woody trees; Lack true flowers.

Angiospermophyta: Bear enclosed seeds inside fruits; Produce true flowers; Have broad leaves; Possess double fertilization.

(b) Differences Table **(3 Marks):** Give 1 mark per correct row (Max 3 marks).

(c) Economic Importance (4 Marks):

Feature	**Coniferophyta**	**Angiospermophyta**
Reproductive Structure	Cones (Strobili)	Flowers
Seed Protection	Naked seeds (not enclosed)	Enclosed within a fruit
Xylem Vessels	Possess tracheids only	Possess true vessels and tracheids

(i) Food supply (grains, fruits, vegetables).

(ii) Medical source (herbal and pharmaceutical products).

(iii) Raw materials (cotton for textiles, timber for construction/paper). *(2 marks per well-explained point)*

Conclusion: **(1 mark)**

Question 10:

(a) **Reasons for Locomotion (6 Marks):**

(i) Searching for food/water to prevent starvation.

(ii) Escaping from predators or harsh environmental conditions (danger).

(iii) Finding mates for reproduction to ensure species continuity.

(b) **Skeleton Parts (3 Marks):**

(i) *Axial Skeleton*: Skull, Ribcage, Vertebral column. *(1 mark for part, 1 mark for any 2 bones)*

(ii) *Appendicular Skeleton*: Limb bones (Humerus, Femur), Pectoral girdle, Pelvic girdle. *(1 mark for part, 1 mark for any 2 bones)*

(c) **Functions of Skeleton (6 Marks):**

(i) Protection of delicate internal organs (e.g., skull protects brain).

(ii) Production of blood cells inside the red bone marrow.

(iii) Storage of essential minerals like calcium and phosphorus.

Question 11:

(a) **Ecosystem Components (5 Marks):**

Definition: A community of living organisms interacting with non-living components of their environment.

Biotic: Plants, Animals, Fungi, Bacteria.

Abiotic: Sunlight, Temperature, Water

(b) **Human Disruptions (6 Marks):**

(i) *Deforestation*: Destroys habitats, causes soil erosion, disrupts the carbon cycle.

(ii) *Industrial/Agricultural Pollution*: Introduces toxic chemicals into water bodies (eutrophication) and air.

(iii) *Overexploitation*: Overhunting or overfishing reduces biodiversity and collapses food webs. *(2 marks per well-explained point)*

(c) **Sustainable Conservation Methods (4 Marks):**

(i) Practicing afforestation and re-afforestation to restore forest cover.

(ii) Establishing protected reserves/national parks and enforcing strict environmental laws against poaching and illegal logging. *(2 marks per well-explained point)*